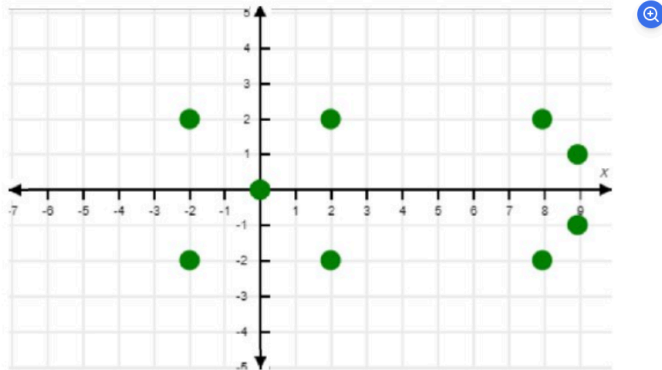


Practice Questions from Week 3

Question 9 : 640653738038

Total Mark : 0.00 | Type : COMPREHENSION

Consider the data points shown in the following image: Based on the above data, answer the given subquestions.



[Discussions \(0\)](#)



Question 10 : 640653738039

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : MCQ

Perform K-means clustering with $K = 2$ and initial cluster centers at $(0, 0)$ and $(6, 0)$. What are the final means of clusters after convergence?

OPTIONS :

- ☐ (2, 0) and (8, 0)
- ☐ (2, 0) and (8.5, 0)
- ☐ (0, 0) and (8, 0)
- ☐ (0, 0) and (8.5, 0)
- ☐ None of these

Solution

Given,

$$\mu_1^0 = (0, 0) \text{ and } \mu_2^0 = (6, 0)$$

$$t = 1$$

x	$\ x - \mu_1^0\ _2^2$		$\ x - \mu_2^0\ _2^2$	z_i
(-2, 2)	$= (-2 - 0)^2 + (2 - 0)^2$ $= 4 + 4 = 8$	<	$= (-2 - 6)^2 + (2 - 0)^2$ $= 64 + 4 = 68$	c_1
(-2, -2)	8	<	68	c_1
(2, -2)	8	<	20	c_1
(2, 2)	8	<	20	c_1
(0, 0)	0	<	36	c_1
(8, 2)	68	>	8	c_2
(9, 1)	82	>	10	c_2
(9, -1)	82	>	10	c_2
(8, -2)	68	>	8	c_2

$$\therefore \mu_1^1 = \left(\frac{-2 + (-2) + 2 + 2 + 0}{5}, \frac{2 + (-2) + (-2) + 2 + 0}{5} \right) = (0, 0)$$

$$\mu_2^1 = \left(\frac{8 + 9 + 9 + 8}{4}, \frac{2 + 1 + (-1) + (-2)}{4} \right) = \left(\frac{34}{4}, \frac{0}{4} \right) = (8.5, 0)$$

$$t = 2$$

$\mathbf{x}$	$\ x - \mu_1^1\ _2^2$		$\ x - \mu_2^1\ _2^2$	z_i
(-2 , 2)	$= (-2 - 0)^2 + (2 - 0)^2$ $= 4 + 4 = 8$	<	$= (-2 - 8.5)^2 + (2 - 0)^2$ $= 110.25 + 4 = 114.25$	c_1
(-2 , -2)	8	<	114.25	c_1
(2 , -2)	8	<	46.25	c_1
(2 , 2)	8	<	46.25	c_1
(0 , 0)	0	<	72.25	c_1
(8 , 2)	68	>	4.25	c_2
(9 , 1)	82	>	1.25	c_2
(9 , -1)	82	>	1.25	c_2
(8 , -2)	68	>	4.25	c_2

$$\therefore \mu_1^2 = \left(\frac{-2 + (-2) + 2 + 2 + 0}{5}, \frac{2 + (-2) + (-2) + 2 + 0}{5} \right) = (0, 0)$$

$$\mu_2^2 = \left(\frac{8 + 9 + 9 + 8}{4}, \frac{2 + 1 + (-1) + (-2)}{4} \right) = \left(\frac{34}{4}, \frac{0}{4} \right) = (8.5, 0)$$

$\therefore$ At $t = 1$ and $t = 2$, the value of μ_1^t and μ_2^t are same.
we can conclude that the algorithm has converged.

Question 11 : 640653738040[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : SA

After introducing a new data point $(7, 0)$, the cluster centers were updated. Enter the sum of the updated x-coordinates for both cluster centers accurately, rounding your answer to two decimal places.

Answer (Numeric):

Solution

$x = (7, 0)$ is added in the dataset

$$d((7, 0), (0, 0)) = \|x - \mu_1\|_2^2 = 49$$

$$d((7, 0), (8.5, 0)) = \|x - \mu_2\|_2^2 = 2.25$$

$\therefore$ the datapoint will be assigned to cluster 2

No changes will be done in μ_1

$$\begin{aligned} \text{The updated } \mu_2 \text{ will be} &= \left(\frac{8 + 9 + 9 + 8 + 7}{5}, \frac{2 + 1 + (-1) + (-2) + 0}{5} \right) \\ &= \left(\frac{41}{5}, 0 \right) = (8.2, 0) \end{aligned}$$

Sum of x-coordinates of the final cluster centers are: $0 + 8.20 = 8.20$

Question 15 : 640653738037

[View Solutions \(0\)](#)

Total Mark : 5.00 | Type : MSQ

Consider Lloyd's algorithm used for k-means clustering and choose the correct statements:

OPTIONS :

- ☐ K-means algorithm may get stuck at local minima.
- ☐ It guarantees finding the optimal clustering (global minimum) in every run.
- ☐ In practice, k should be as large as possible.
- ☐ If the resources are limited and the data set is huge, it will be good to prefer K-means over K-means++.

Solution

- a) True – K-means can get stuck in local minima due to poor initial centroid selection.
- b) False – K-means does not guarantee finding the global minimum in every run.
- b) False – A very large k can lead to overfitting and unnecessary computation.
- d) True – K-means is preferable over K-means++ when resources are limited, as K-means++ has higher initial computation costs.

Question 8 : 640653827438

[View Solutions \(0\)](#) Total Mark : 4.00 | Type : SA

Consider the k-means++ algorithm with $k = 2$ for a dataset that has 7 points:
 $D = \{x_1, \dots, x_7\}$. The data-point x_7 has been chosen as the first mean, that is, $\mu_1 = x_7$.
The Euclidean distance of the remaining points from this mean is given below:

Point	$d(x_i, \mu_1)$
x_1	1
x_2	2
x_3	$\sqrt{2}$
x_4	$\sqrt{3}$
x_5	$\sqrt{5}$
x_6	1

What is the probability of choosing x_2 as the second mean given that x_7 has been chosen as the first mean?

Answer (Numeric):

Answer

Solution

Point	$d(x_i, \mu_1)$	$d(x_i, \mu_1)^2$
x_1	1	1
x_2	2	4
x_3	$\sqrt{2}$	2
x_4	$\sqrt{3}$	3
x_5	$\sqrt{5}$	5
x_6	1	1

$$p(x_2) = \frac{d(x_2, \mu_1)^2}{\sum_{i=1}^n d(x_i, \mu_1)^2} = \frac{4}{16} = 0.25$$

Question 13 : 640653827444

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the following dataset of six points in $\mathbb{R}^2$:



x	y
-1	-4
5	6
-2	-2
4	5
-3	-3
3	4

K-means clustering is run on this dataset with $k = 2$. In this version, the means are initialized first. The mean of the first cluster is initialized to $(-4, -4)$ and the mean of the second cluster is initialized to $(6, 6)$.

[Discussions \(0\)](#)**Question 14 : 640653827445**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If (x_1, y_1) is the mean of the first cluster after convergence, find the value of $x_1 + y_1$.

Answer (Numeric):

[Check Answer](#)[Discussions \(0\)](#)**Question 15 : 640653827446**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If (x_2, y_2) is the mean of the second cluster after convergence, find the value of $x_2 + y_2$.

Answer (Numeric):

[Check Answer](#)

Solution

Given, $\mu_1^0 = (-4, -4)$ and $\mu_2^0 = (6, 6)$

$t = 1$

x	$\ x - \mu_1^0\ _2^2$		$\ x - \mu_2^0\ _2^2$	z_i
(-1, -4)	9	<	149	c_1
(5, 6)	181	>	1	c_2
(-2, -2)	8	<	128	c_1
(4, 5)	145	>	5	c_2
(-3, -3)	2	<	162	c_1
(3, 4)	113	>	13	c_2

$$\mu_1^1 = \left(\frac{-1 + (-2) + (-3)}{3}, \frac{-4 + (-2) + (-3)}{3} \right) = (-2, -3)$$

$$\mu_2^1 = \left(\frac{5 + 4 + 3}{3}, \frac{6 + 5 + 4}{3} \right) = (4, 5)$$

$$t = 2$$

$\mathbf{x}$	$\ x - \mu_1^1\ _2^2$		$\ x - \mu_2^1\ _2^2$	z_i
(-1 , -4)	2	<	106	c_1
(5 , 6)	130	>	2	c_2
(-2 , -2)	1	<	85	c_1
(4 , 5)	100	>	0	c_2
(-3 , -3)	1	<	113	c_1
(3 , 4)	74	>	2	c_2

$$\mu_1^2 = \left(\frac{-1 + (-2) + (-3)}{3}, \frac{-4 + (-2) + (-3)}{3} \right) = (-2, -3)$$

$$\mu_2^2 = \left(\frac{5 + 4 + 3}{3}, \frac{6 + 5 + 4}{3} \right) = (4, 5)$$

$\therefore$ At $t = 1$ and $t = 2$, the value of μ_1^t and μ_2^t are same.

we can conclude that the algorithm has converged.

$$x_1 + y_1 = -2 + (-3) = -5$$

$$x_2 + y_2 = 4 + 5 = 9$$

Question 8 : 640653992024[View Solutions \(0\)](#)

Total Mark : 5.00 | Type : SA

Consider the following data points for k-means clustering.

$(-1, 0), (-1, 1), (-1, -1), (2, 0), (3, 1), (3, -1), (4, 0)$

In the initialization step of k-means with $k = 2$, suppose $\mu_1^0 = (-1, 0)$ and $\mu_2^0 = (2, 0)$.

Distances of datapoints from initial cluster means is tabulated below:

x_i	$\ x_i - \mu_1^0\ _2^2$	$\ x_i - \mu_2^0\ _2^2$
$(-1, 0)$	0	3
$(-1, 1)$	1	10
$(-1, -1)$	1	10
$(2, 0)$	3	0
$(3, 1)$	17	2
$(3, -1)$	17	2
$(4, 0)$	5	2

As per these cluster centers, the data points are then assigned to either cluster 1 or cluster 2. After this assignment, what will be the value of the objective function?

Note: Objective function is given by

$$F(z_1, z_2, \dots, z_n) = \sum_{i=1}^n \|x_i - \mu_{z_i}\|_2^2$$

Answer (Numeric):

Solution

Given $\mu_1^0 = (-1, 0)$ and $\mu_2^0 = (2, 0)$

$t = 1$

x	$\ x - \mu_1^0\ _2^2$	$\ x - \mu_2^0\ _2^2$	z_i
(-1, 0)	0	3	1
(-1, 1)	1	10	1
(-1, -1)	1	10	1
(2, 0)	3	0	2
(3, 1)	17	2	2
(3, -1)	17	2	2
(4, 0)	25	2	2

$\therefore \mu_1^1 = (-1, 0)$ and $\mu_2^1 = (3, 0)$

$$t = 2$$

$\mathbf{x}$	$\ x - \mu_1^1\ _2^2$	$\ x - \mu_2^1\ _2^2$	z_i
(-1 , 0)	0	16	1
(-1 , 1)	1	17	1
(-1 , -1)	1	17	1
(2 , 0)	9	1	2
(3 , 1)	17	1	2
(3 , -1)	17	1	2
(4 , 0)	25	1	2

$$\therefore \mu_1^2 = (-1, 0) \text{ and } \mu_2^2 = (3, 0)$$

$\therefore$ At $t = 1$ and $t = 2$, the value of μ_1^t and μ_2^t are same.

we can conclude that the algorithm has converged.

$$\therefore F(Z) = 1 + 1 + 1 + 1 + 1 + 1 = 6$$

Question 14 : 640653992021

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

A k-means++ algorithm with $k = 3$ is applied on the following 2D points:

$(0, 1), (1, 0), (1, 2), (2, 1), (2, 3), (2, 4), (3, 2)$

First cluster mean μ_1^0 is chosen as $(2, 1)$.

Suppose the point with the highest score is chosen as the 2nd cluster mean μ_2^0 .

[Discussions \(0\)](#)**Question 15 : 640653992022**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : MCQ

What is μ_2^0 ? Use squared distance to calculate the scores.

OPTIONS :

☐ (0,1)

☐ (2,3)

☐ (3,2)

☐ (2,4)

Solution

Given, $\mu_1^0 = (2, 1)$

metric to select μ_2^0 is data point with the highest score

formulae to calculate score = $\|x - \mu_1\|_2^2$

$$(0, 1) : (0 - 2)^2 + (1 - 1)^2 = 4 + 0 = 4$$

$$(1, 0) : (1 - 2)^2 + (0 - 1)^2 = 1 + 1 = 2$$

$$(1, 2) : (1 - 2)^2 + (2 - 1)^2 = 1 + 1 = 2$$

$$(2, 1) : (2 - 2)^2 + (1 - 1)^2 = 0 + 0 = 0$$

$$(2, 3) : (2 - 2)^2 + (3 - 1)^2 = 0 + 4 = 4$$

$$(2, 4) : (2 - 2)^2 + (4 - 1)^2 = 0 + 9 = 9$$

$$(3, 2) : (3 - 2)^2 + (2 - 1)^2 = 1 + 1 = 2$$

Highest squared distance is 9, corresponding to the point (2, 4)

$$\therefore \mu_2^0 = (2, 4)$$

Question 16 : 640653992023[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : MCQ

Which point has the lowest probability of being chosen as the 3rd cluster mean? Use squared distance to calculate the scores.

OPTIONS :

☐ (1,0)☐ (2,3)☐ (3,2)☐ (1,2)**Solution**

Given, $\mu_1^0 = (2, 1)$ and $\mu_2^0 = (2, 4)$

$\mathbf{x}$	$\ x - \mu_1\ _2^2$	$\ x - \mu_2\ _2^2$	$(\min d(x, \mu_1)^2, d(x, \mu_2)^2)$	$P(x)$
(0 , 1)	4	13	4	$= \frac{4}{11}$
(1 , 0)	2	17	2	$= \frac{2}{11}$
(1 , 2)	2	5	2	$= \frac{2}{11}$
(2 , 1)	0	9	0	$= 0$
(2 , 3)	4	1	1	$= \frac{1}{11}$
(2 , 4)	9	0	0	$= 0$
(3 , 2)	2	5	2	$= \frac{2}{11}$

$\therefore p(x_5)$ is lowest

$\therefore$ It has lowest chance of getting selected as the next cluster mean.